

1 This question is about states of matter.

(a) Complete the table, using ticks (✓) and crosses (✗), to describe the properties of gases, liquids and solids.

state of matter	particles are touching	particles have random movement	particles are regularly arranged
gas			
liquid			
solid			

[3]

(b) Substances can change state.

(i) Boiling and evaporation are two ways in which a liquid changes into a gas.

Describe **two** differences between boiling and evaporation.

1

2 [2]

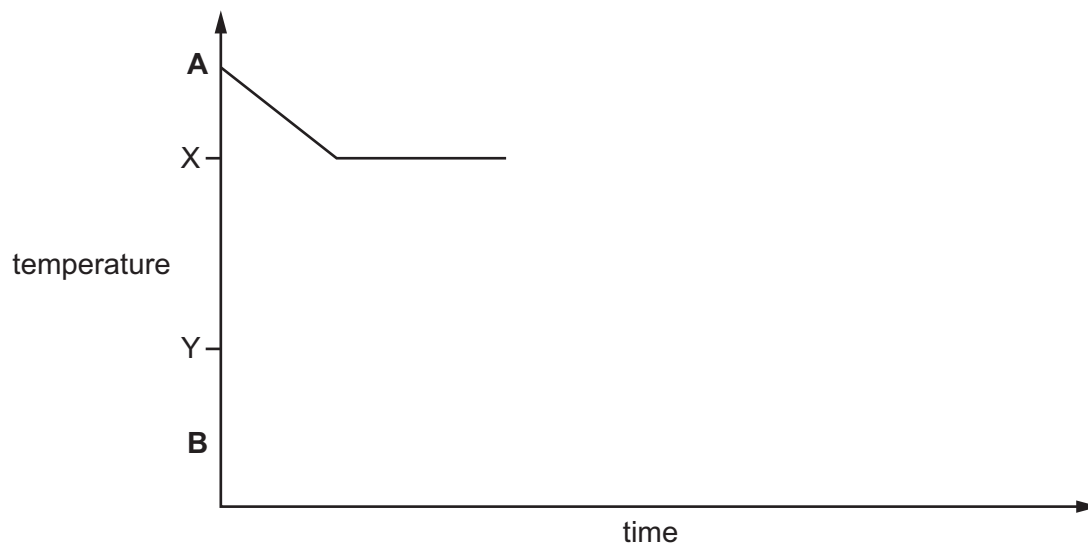
(ii) Name the change of state when:

• a gas becomes a liquid

• a solid becomes a gas. [2]

- (c) A substance boils at temperature X and melts at temperature Y.

Complete the graph to show the change in temperature over time as the substance cools from temperature A to temperature B.



[2]

- (d) A solution is a mixture of a solute and a solvent.

(i) Name the process when a solid substance mixes with a solvent to form a solution.

..... [1]

(ii) Name the type of reaction when two solutions react to form an insoluble substance.

..... [1]

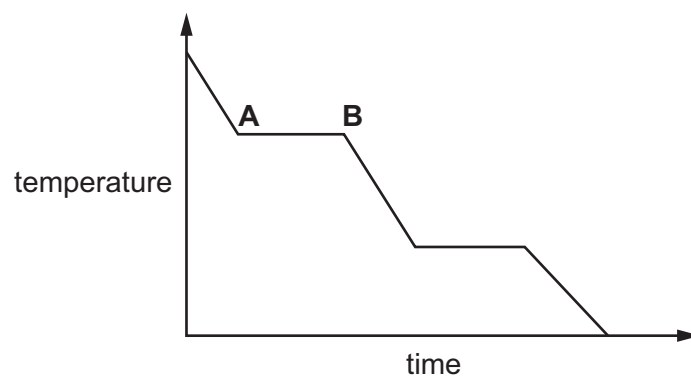
[Total: 11]

5 (a) Complete the table about solids, liquids and gases.

	particle separation	particle arrangement	type of motion
solid		regular	vibrate only
liquid	touching		random
gas	apart	random	

[3]

(b) The graph shows the change in temperature as a sample of a gas is cooled.



Name the change of state taking place between **A** and **B**.

..... [1]

(c) A bottle of liquid perfume is left open at the front of a room.

After some time, the perfume is smelt at the back of the room.

Name the **two** physical processes taking place.

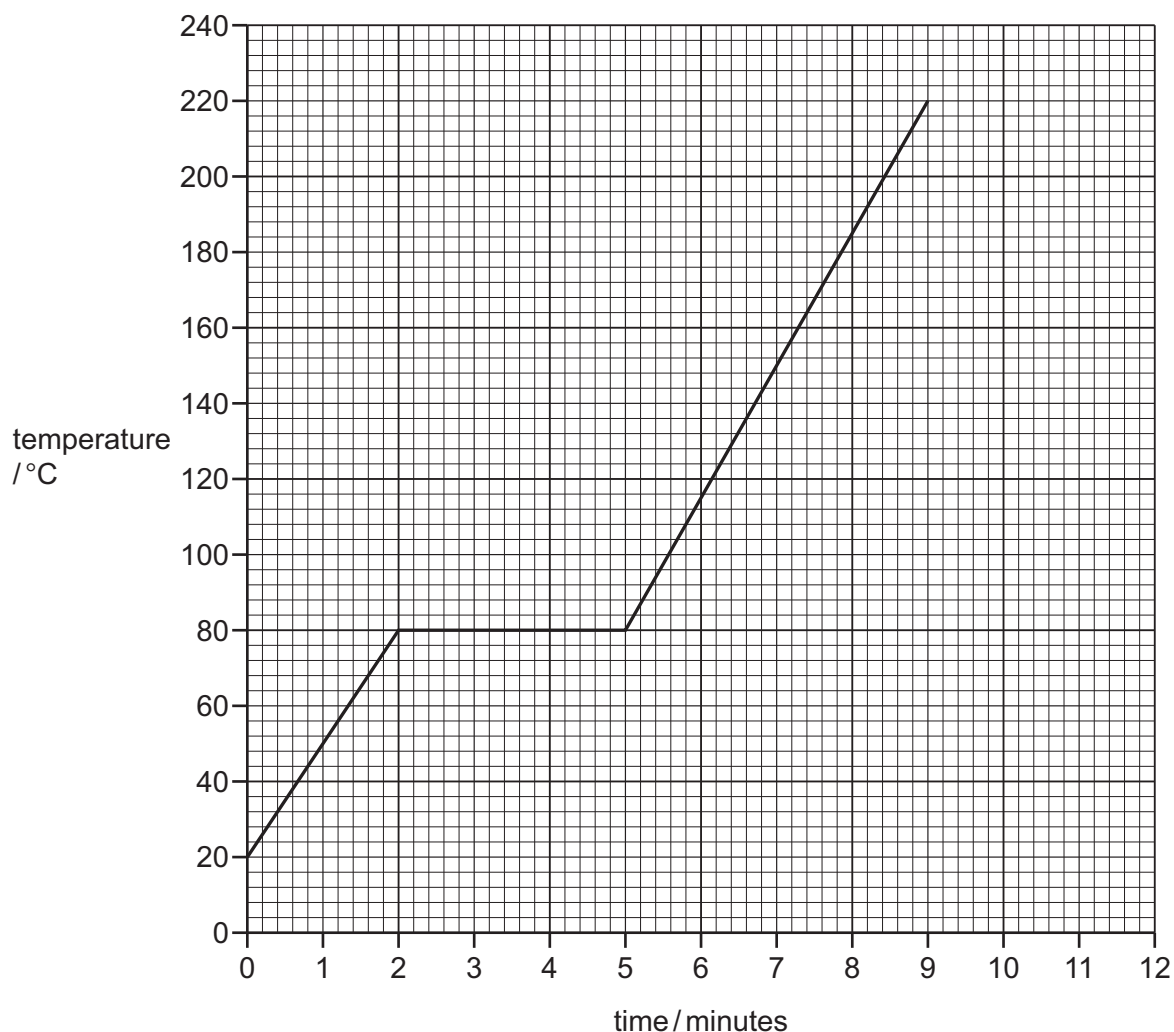
1

2 [2]

[Total: 6]

- 2 **Z** is a covalent substance. In an experiment, a sample of pure solid **Z** was continually heated for 11 minutes.

The graph shows how the temperature of the sample of pure **Z** changed during the first 9 minutes.



- (a) What is the melting point of pure **Z**?

..... °C [1]

- (b) The sample of pure **Z** began to boil at 9 minutes. It was boiled for 2 minutes.

Use this information to sketch on the grid how the temperature of the sample of pure **Z** changed between 9 minutes and 11 minutes. [1]

- (c) The sample of pure **Z** was continually heated between 2 minutes and 5 minutes.

Explain, in terms of attractive forces, why there was no increase in the temperature of the sample of pure **Z** between 2 minutes and 5 minutes.

.....

 [2]

- (d) Describe how the motion of particles of pure **Z** changed from 0 minutes to 2 minutes.

.....
 [2]

- (e) The experiment was repeated using a solid sample of **impure Z**.

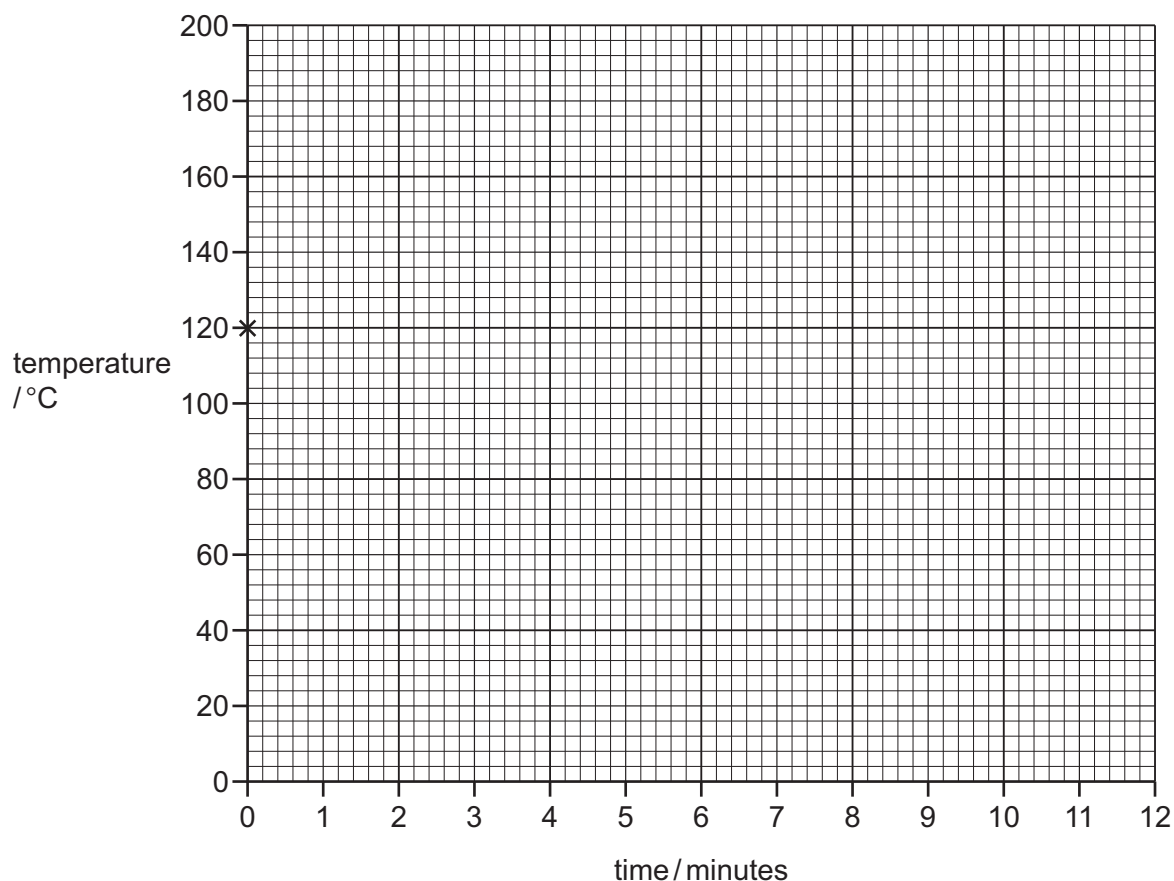
Suggest the differences, if any, in the melting point and boiling point of the sample of impure **Z** compared to the sample of pure **Z**.

melting point

boiling point [2]

- (f) A sample of pure **Z** was allowed to cool from 120 °C to 20 °C. The total time taken was 8 minutes.

Starting from point **x**, sketch on the grid how the temperature of the sample of pure **Z** changed between 0 minutes and 8 minutes.

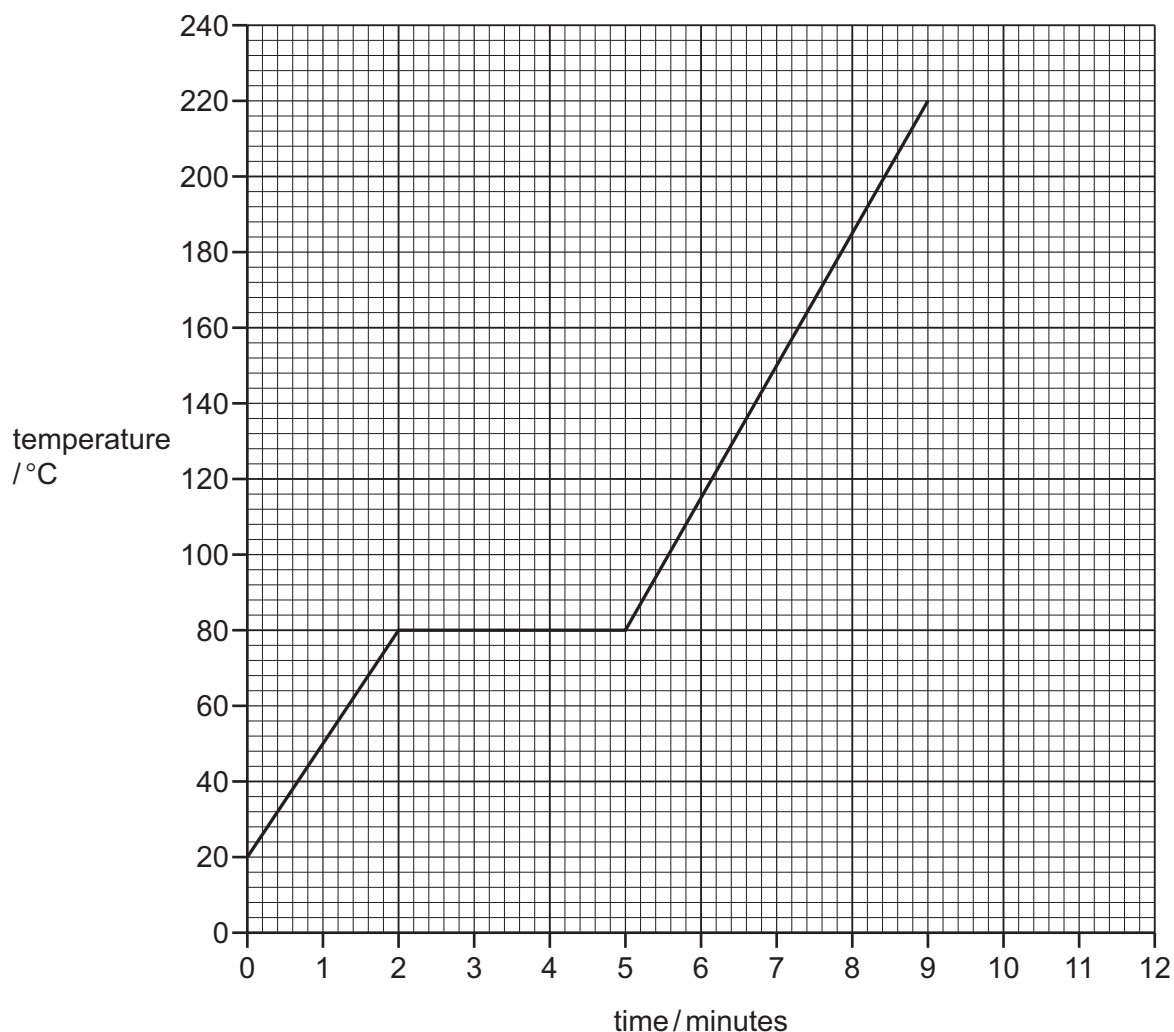


[2]

[Total: 10]

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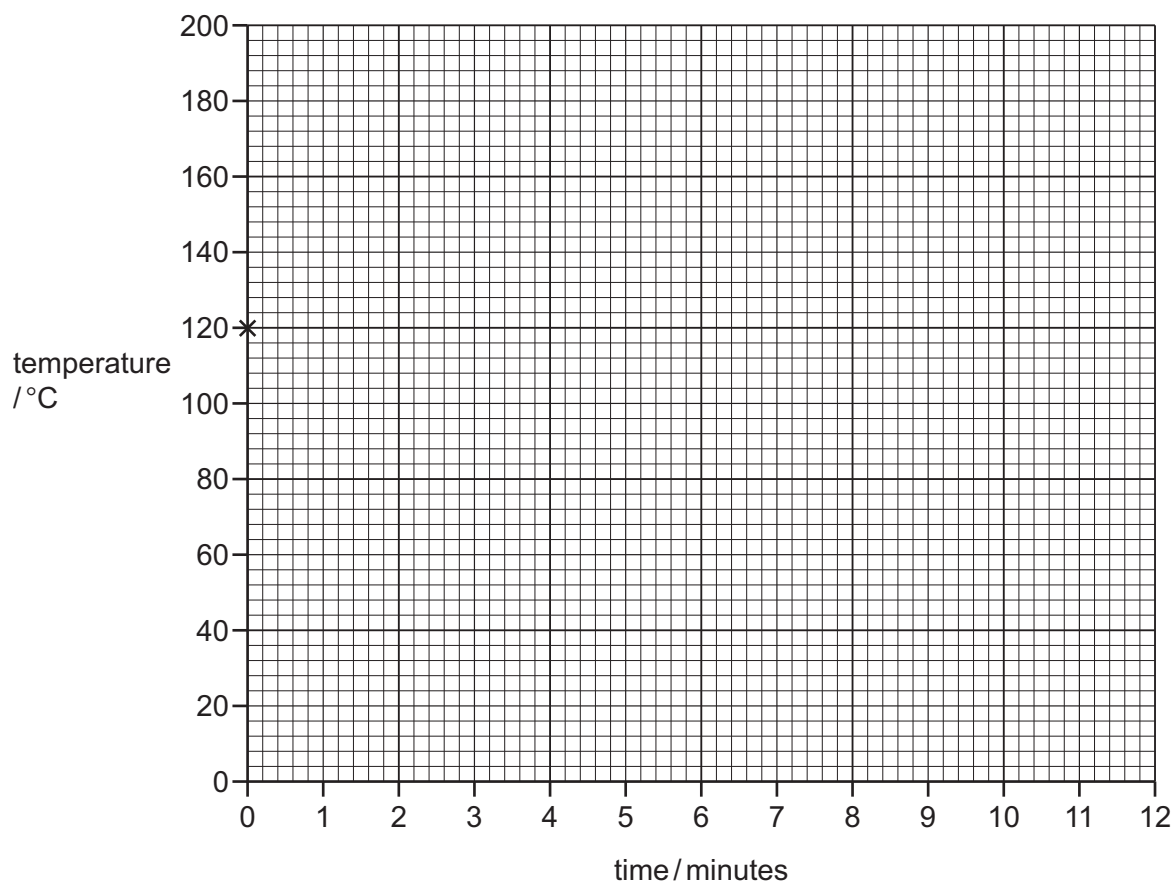
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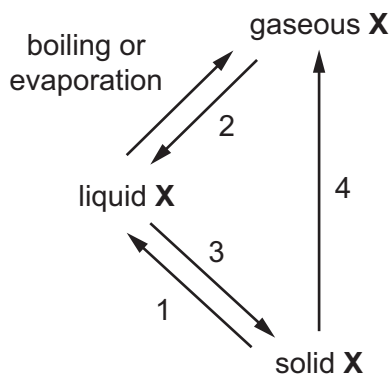
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[2]

[Total: 10]

- 1 Element **X** can undergo the following physical changes.



- (a) (i) Give the scientific name for each of the numbered physical changes.

1 [4]

2

3

4

- (ii) Explain why the changes shown are physical changes.

..... [1]

.....

- (iii) One difference between boiling and evaporation is the rate at which the processes occur.
State **one** other difference between boiling and evaporation.

..... [1]

.....

- (b) Describe the separation, arrangement and motion of particles of element **X** in the solid state.

separation

arrangement

motion [3]

- (c) Element **X** is a Group I metal. It burns in air to form an oxide X_2O .

Write a chemical equation for this reaction.

..... [2]

[Total: 11]

1 (a) Dust particles in the air move around in a random way.

(i) What term describes the random movement of the dust particles?

..... [1]

(ii) Identify the particles in the air which cause the random movement of the dust particles.

..... [2]

(iii) Explain why the dust particles move in this way.

.....

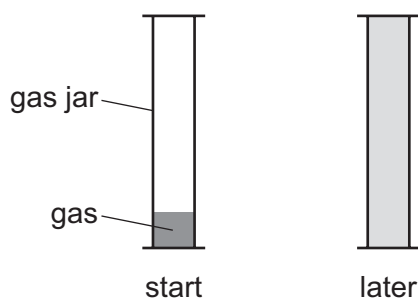
.....

..... [2]

(b) When chlorine gas, Cl_2 , is put into a gas jar, it spreads out to fill the gas jar.

When bromine gas, Br_2 , is put into a gas jar, it also spreads out to fill the gas jar.

The process takes longer for bromine gas than for chlorine gas.



(i) What term describes the way that the gas particles spread out?

..... [1]

(ii) Use **data** from the Periodic Table to explain why bromine gas takes longer to fill a gas jar than chlorine gas.

.....

.....

..... [2]

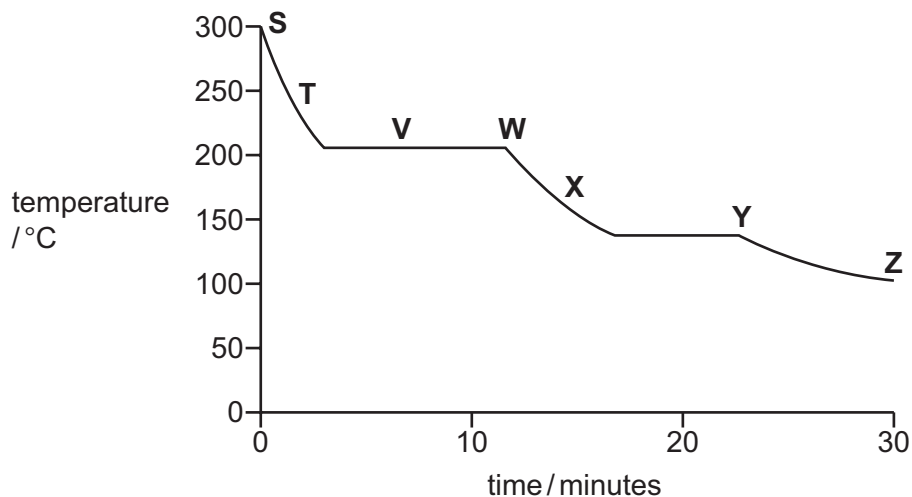
(iii) Explain why increasing the temperature increases the rate at which the gas particles spread out.

.....

..... [1]

[Total: 9]

- 2 The graph shows how the temperature of a substance changes as it is cooled over a period of 30 minutes. The substance is a gas at the start.



Each letter on the graph may be used once, more than once or not at all.

- (a) Which letter, **S**, **T**, **V**, **W**, **X**, **Y** or **Z**, shows when

(i) the particles in the substance have the most kinetic energy,

..... [1]

(ii) the particles in the substance are furthest apart,

..... [1]

(iii) the substance exists as both a gas and a liquid?

..... [1]

- (b) Use the graph to estimate the freezing point of the substance.

..... °C [1]

- (c) Name the change of state directly from a solid to a gas.

..... [1]

- (d) When smoke is viewed through a microscope, the smoke particles in the air appear to jump around.

(i) What term describes this movement of the smoke particles?

..... [1]

(ii) Explain why the smoke particles move in this way.

.....

.....

..... [2]

[Total: 8]

1 Particles behave differently when in different physical states.

- (a)** Solids have a fixed volume and a definite shape.
Gases have no fixed volume and take the shape of the container.

Describe the volume and shape of liquids.

.....
..... [1]

- (b)** Complete the table to show the separation, arrangement and movement of particles in each physical state.

state	separation of particles	arrangement of particles	movement of particles
solid			
liquid	touching one another	randomly arranged	move over one another
gas			

[6]

- (c)** Name the following changes of state.

- (i)** Ice turning into water.

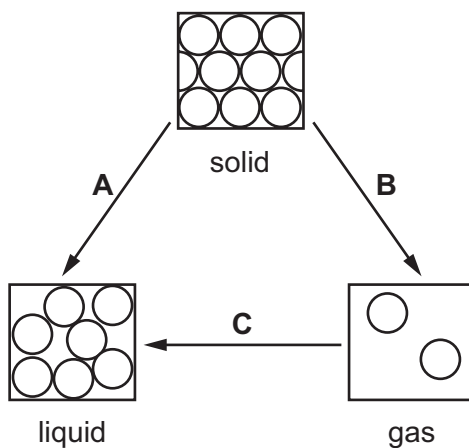
..... [1]

- (ii)** Solid carbon dioxide turning directly into gaseous carbon dioxide at room temperature.

..... [1]

[Total: 9]

2 Matter can exist as solid, liquid or gas. The arrows show some changes of state.



(a) Name the changes of state represented on the diagram.

(i) **A** [1]

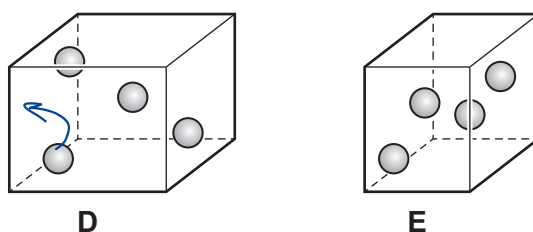
(ii) **B** [1]

(iii) **C** [1]

(b) Explain why energy has to be supplied to turn a liquid into a gas.

.....
 [1]

(c) The diagrams represent the same number of particles of a gas in two containers, **D** and **E**, which have different volumes. The two containers are at the same temperature.



In which container will the pressure be higher? Explain your answer.

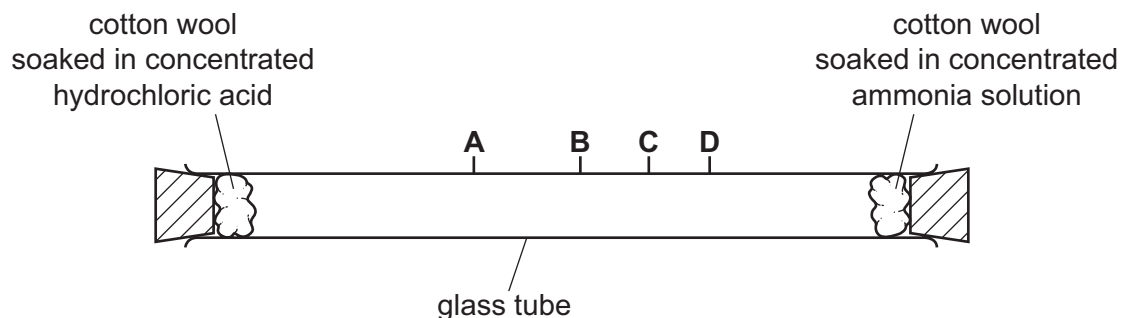
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 [1]

[Total: 5]

- 6 Concentrated ammonia solution gives off ammonia gas. Concentrated hydrochloric acid gives off hydrogen chloride gas. Ammonia, NH_3 , and hydrogen chloride, HCl , are both colourless gases. Ammonia reacts with hydrogen chloride to make the white solid ammonium chloride.

Apparatus is set up as shown.



After ten minutes a white solid forms in the tube where the gases meet.

- (a) (i) Write the chemical equation for the reaction of ammonia with hydrogen chloride.

..... [1]

- (ii) Name the process by which the ammonia and hydrogen chloride gases move in the tube.

..... [1]

- (iii) At which point, **A**, **B**, **C** or **D**, does the white solid form? Explain why the white solid forms at that point.

the solid forms at

explanation

..... [3]

- (iv) The experiment was repeated at a higher temperature.

Predict how the results of the experiment would be different. Explain your answer.

.....

.....

..... [3]